Linux Basics for Hackers

CHAPTER-TO-CHALLENGE MAP

A companion guide mapping each chapter of *Linux Basics for Hackers* by OccupyTheWeb to hands-on practice on **OverTheWire** and **picoCTF / picoGym**. Read the chapter, then immediately drill it on the matching challenge — that's the loop that makes this stick.

picoGym note: picoCTF's practice site reuses problems from past competitions, but exact challenge titles get added/retired each year. This guide points to the **category to filter by** (General Skills, Cryptography, Forensics, Web Exploitation, Reverse Engineering, Binary Exploitation) plus what to search for in the difficulty filter.

Chapters 1–3: Foundations

The first three chapters build your terminal fluency — navigation, text processing, and network awareness. Bandit levels 0–13 cover all of this in sequence.

CHAPTER 1

Getting Started with the Basics

Topic: Navigation, core commands

- **OverTheWire:** Bandit 0–6 — ls, cd, cat, find, hidden files
- **picoGym:** General Skills, easiest tier — basic Linux/terminal warmups

CHAPTER 2

Text Manipulation

Topic: grep, sed, awk, cut, sort

- **OverTheWire:** Bandit 7–13 — heavy grep/sort/uniq use to extract passwords from noisy files
- **picoGym:** General Skills / Forensics — challenges asking you to extract a flag from a large text/log dump

CHAPTER 3

Analyzing and Managing Networks

Topic: netstat, ifconfig/ip, nmap, ports

- **OverTheWire:** Bandit 24, 27 (netcat-driven levels) + try Leviathan for service interaction
- **picoGym:** General Skills — networking-tagged problems; also Forensics challenges involving packet captures (.pcap files)

Chapters 4–6: Permissions, Processes & Environment

Adding and Removing Software

Topic: apt/dpkg, package management

- **OverTheWire:** No direct OTW equivalent — best practiced on your own VM. Use Bandit as a sandbox to inspect installed binaries with `dpkg -l` once connected
- **picoGym:** Not heavily tested directly; skip ahead, revisit when setting up your own attack VM

CHAPTER 5

Controlling File and Directory Permissions

Topic: chmod, chown, SUID/SGID, ACLs

- **OverTheWire:** Bandit 14, 15, 25, 31–33 — permission-gated files; Leviathan (almost entirely about exploiting SUID binaries)
- **picoGym:** General Skills — "permissions" or "SUID" tagged challenges in the easy tier

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CHAPTER 6

Process Management

Topic: ps, kill, top, background jobs

- **OverTheWire:** Bandit 32–33 touch on this; deeper practice: Leviathan levels involve inspecting running binaries
- **picoGym:** Binary Exploitation (intro tier) — challenges that have you interact with a running process to extract a flag

Chapters 7–9: Scripting, Archives & Environment

Chapter 7 — Managing User Environment Variables

Topic: $PATH, export, shell config files

- **OverTheWire:** Bandit 27–28 (git/environment quirks); look for "PATH hijacking" style levels in Narnia
- **picoGym:** General Skills — environment/PATH manipulation challenges, often tagged "easy"

Chapter 8 — Bash Scripting

Topic: Writing your own scripts/loops/conditionals

- **OverTheWire:** Bandit 33 and any level requiring you to script a brute-force or automate a repetitive task (several late Bandit levels reward scripting over manual repetition)
- **picoGym:** General Skills — challenges that explicitly require scripting (e.g., generating/brute-forcing many values)

Chapter 9 — Compressing and Archiving

Topic: tar, gzip, zip, nested archives

- **OverTheWire:** Bandit 8–11 — multiple nested compression formats stacked together
- **picoGym:** Forensics — "extract the flag from this archive" style challenges, often multi-layered

Chapters 10–12: Systems, Logging & Services

1

Chapter 10 — Filesystem and Storage Device Management

Topic: Mounting, partitions, /proc, /sys, fdisk

- **OverTheWire:** No strong OTW match — practice on your own VM by mounting a second disk/USB and exploring /proc
- **picoGym:** Forensics — disk image analysis challenges (.img/.dd files) are the closest fit

2

Chapter 11 — The Logging System

Topic: /var/log, journalctl, syslog

- **OverTheWire:** Bandit doesn't focus on this directly; check /var/log access patterns once inside any Bandit level as a side-exercise
- **picoGym:** Forensics — log analysis challenges (auth.log, access.log style problems)

3

Chapter 12 — Using and Abusing Services

Topic: Exploiting running daemons, weak service configs

- **OverTheWire:** Leviathan and Narnia — both built around exploiting locally running services/binaries
- **picoGym:** Web Exploitation (if web service) or Binary Exploitation (if local daemon) — depends which service angle you're drilling

Chapters 13–14: OPSEC & Wireless

⚠️ WORKSHOP-ONLY CHAPTERS

These two chapters are the **exceptions** in this guide — they cover OPSEC, anonymity tooling, and wireless hardware that don't map to puzzle-style CTF challenges. Treat them as **standalone workshop sessions** rather than assigned homework.

CHAPTER 13

Becoming Secure and Anonymous

Topic: VPNs, Tor, OPSEC, anonymity tooling

- **OverTheWire:** Not OTW's focus — conceptual/tooling practice outside wargames
- **picoGym:** Doesn't test this directly either

Treat as a **club discussion/workshop topic** rather than a CTF skill. Great for a "Tor + OPSEC discussion night."

CHAPTER 14

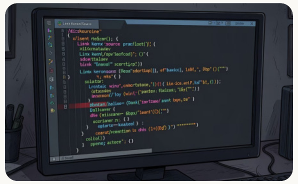
Understanding and Inspecting Wireless Networks

Topic: Wi-Fi monitor mode, packet capture, deauth

- **OverTheWire:** No OTW equivalent (needs real RF hardware)
- **picoGym:** picoCTF doesn't cover wireless either

This is the one chapter where you'll want a **separate hands-on lab** — e.g., a club Wi-Fi monitor-mode session on hardware you own.

Chapters 15–17: Kernel, Cron & Python



Chapter 15 — Managing the Linux Kernel and Loadable Kernel Modules

Topic: Kernel modules, lsmod, insmod

- **OverTheWire:** Behemoth / Utumno start to touch kernel-adjacent exploitation once you're past basic binary exploitation
- **picoGym:** Binary Exploitation — harder tier, kernel-adjacent challenges are rarer but tagged when present

Chapter 16 — Automating Tasks with Job Scheduling

Topic: cron, at, scheduled task abuse

- **OverTheWire:** Bandit 21, 27 — cron-based privilege escalation is a classic Bandit lesson
- **picoGym:** General Skills — privilege escalation via cron is a recurring theme in easy/medium tier

Chapter 17 — Python Scripting Basics for Hackers

Topic: Writing exploit/automation scripts in Python

- **OverTheWire:** Use Python to automate brute-forcing or pattern-matching across any late Bandit level, or write a solver for Krypton's cipher levels
- **picoGym:** Cryptography and Reverse Engineering — most challenges in these categories expect a short Python script to solve, making this the best category to pair with the chapter

Full Chapter Reference Table

Use this table as a quick lookup when planning sessions or assigning homework.

Ch.	Topic	OverTheWire	picoGym Category
1	Navigation, core commands	Bandit 0–6	General Skills, easiest tier
2	grep, sed, awk, cut, sort	Bandit 7–13	General Skills / Forensics
3	netstat, ifconfig, nmap, ports	Bandit 24, 27; Leviathan	General Skills / Forensics (.pcap)
4	apt/dpkg, package management	Own VM practice	— skip, revisit later
5	chmod, chown, SUID/SGID	Bandit 14,15,25,31–33; Leviathan	General Skills (SUID tagged)
6	ps, kill, top, background jobs	Bandit 32–33; Leviathan	Binary Exploitation (intro)
7	$PATH, export, shell config	Bandit 27–28; Narnia	General Skills (PATH/easy)
8	Bash scripting, loops, conditionals	Bandit 33; late Bandit levels	General Skills (scripting)
9	tar, gzip, zip, nested archives	Bandit 8–11	Forensics (archive extraction)
10	Mounting, partitions, /proc	Own VM practice	Forensics (.img/.dd)
11	/var/log, journalctl, syslog	Side-exercise in Bandit	Forensics (log analysis)
12	Exploiting running daemons	Leviathan, Narnia	Web Exploit / Binary Exploit
13	VPNs, Tor, OPSEC	— workshop only	— workshop only
14	Wi-Fi monitor mode, deauth	— hardware lab	— hardware lab
15	Kernel modules, lsmod, insmod	Behemoth / Utumno	Binary Exploitation (hard)
16	cron, at, scheduled task abuse	Bandit 21, 27	General Skills (cron priv esc)
17	Python exploit/automation scripts	Bandit + Krypton solver	Cryptography / Reverse Eng.

How to Use This With the Club

1

Read-then-drill loop

Assign a chapter as reading **before** each session, then spend the session live-solving the matched OverTheWire levels together. The immediate application cements the concepts.

2

picoGym as homework

Since OTW levels are sequential and shared, use picoGym (filtered by the category listed in the table) as **individual homework** so people aren't blocked waiting on each other.

3

Chapters 4, 10, 13, 14 are the exceptions

They're light on OTW/picoCTF matches because they're systems/OPSEC/hardware topics rather than puzzle-style exploitation. Treat those as **standalone workshop sessions** — e.g., "set up a VM and explore your own filesystem," "Tor + OPSEC discussion night" — rather than CTF homework.

4

Chapter 17 (Python) pairs with your branch

Worth pairing with whichever branch (web/binary/crypto) the club ends up choosing from the earlier roadmap — **most real progress past beginner level requires scripting.**

Quick Links

All platforms are free to use. Start with Bandit, then branch out as your skills grow.

Bandit

The primary wargame for Chapters 1–3, 5, 7–9, 16. Start here.

overthewire.org/wargames/bandit/

Leviathan

SUID exploitation and local service abuse — Chapters 5, 6, 12.

overthewire.org/wargames/leviathan/

Narnia

PATH hijacking, environment manipulation, local daemon exploitation — Chapters 7, 12.

overthewire.org/wargames/narnia/

Behemoth

Kernel-adjacent and advanced binary exploitation — Chapter 15.

overthewire.org/wargames/behemoth/

Krypton

Cipher challenges — pair with Chapter 17 (Python scripting solvers).

overthewire.org/wargames/krypton/

picoCTF / picoGym

Category-filtered practice problems for homework. Filter by General Skills, Forensics, Cryptography, Binary Exploitation, Web Exploitation, or Reverse Engineering.

picoctf.org

Remember the loop: Read the chapter → solve the matching OTW levels together → assign picoGym as homework. Repeat for every chapter.